

Assignment 2 Solutions

Fall 2026

1.18

$$x_1 = -5 + 3x_2$$

x_2 is free

$$x_3 = 8$$

$$x_4 = 5$$

$$1.19 \quad \begin{bmatrix} 1 & 0 & -3 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -3 & 0 & 1 \\ 0 & 1 & 1 & 0 & -4 \\ 0 & 0 & 0 & 1 & 5 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - R_3$$

$$x_1 = 1 + 3x_3$$

$$x_2 = -4 - x_3$$

x_3 is free

$$x_4 = 5$$

$$1.26 \quad \begin{bmatrix} 5 & 2 & 9 \\ 7 & 3 & 12 \\ 2 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 7 & 3 & 12 \\ 5 & 2 & 9 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 14 & 6 & 24 \\ 5 & 2 & 9 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 14 & 6 & 24 \\ 10 & 4 & 18 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 3 \\ 10 & 4 & 18 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 3 \\ 0 & -1 & 3 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3$$

$$R_2 \leftarrow 2R_2$$

$$R_3 \leftarrow 2R_3$$

$$R_2 \leftarrow R_2 - 7R_1$$

$$R_3 \leftarrow R_3 - 5R_1$$

$$\sim \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 6 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - R_2 \quad R_2 \leftarrow -R_2 \quad R_1 \leftarrow R_1 - R_2 \quad R_1 \leftarrow \frac{1}{2}R_1$$

$$1.28 \quad \begin{bmatrix} 2 & 3 & -13 & -8 \\ 1 & 1 & -4 & -1 \\ 1 & 0 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -4 & -1 \\ 2 & 3 & -13 & -8 \\ 1 & 0 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -4 & -1 \\ 0 & 1 & -5 & -6 \\ 1 & 0 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -4 & -1 \\ 0 & 1 & -5 & -6 \\ 0 & -1 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -4 & -1 \\ 0 & 1 & -5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_1$$

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_3 \leftarrow R_3 - R_1$$

$$R_3 \leftarrow R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - R_2$$

$$1.31 \begin{bmatrix} 1 & 1 & -1 & -9 \\ 0 & 1 & -2 & -1 \\ -1 & -1 & 0 & -10 \\ 1 & 1 & 0 & -10 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -1 & -9 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & -10 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$x_1 = -7$$

$$x_2 = -3$$

$$x_3 = -1$$

$$-7 + (-3) - (-1) = -9 \checkmark$$

$$-3 - 2(-1) = -1 \checkmark$$

$$-7 + (-3) = -10 \checkmark$$

$$1.35 \begin{bmatrix} -1 & -2 & -1 & -4 \\ 1 & 2 & 0 & 3 \\ -2 & -4 & 1 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 3 \\ -1 & -2 & -1 & -4 \\ -2 & -4 & 1 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

NO SOLUTION

$$1.40 \begin{bmatrix} -4 & -7 & 4 \\ 3 & 8 & 2 \\ -10 & 1 & -9 \end{bmatrix} \sim \begin{bmatrix} 3 & 8 & 2 \\ -4 & -7 & 4 \\ -10 & 1 & -9 \end{bmatrix} \sim \begin{bmatrix} 3 & 8 & 2 \\ -14 & -6 & -5 \\ -10 & 1 & -9 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$R_2 \leftarrow R_2 + R_3$$

1.45 True. A is inconsistent $\Rightarrow$ RREF of A has row $0 \dots 0 \ 1$ and the 1 is a leading entry, so it is a pivot (in the last column)

1.48 True. A free variable can be given any distinct value in a particular solution

1.49 False. The RREF of a matrix is unique, and the RREF determines the pivot positions, not the row operations

$$1.50 \text{ False} \quad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}$$

both echelon forms

1.58 Trx. $\begin{bmatrix} a & 2a \\ b & 3b \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ 1 pivot per row/col.

1.61 a. $\begin{bmatrix} 0 & 4 & 1 & 16 \\ 9 & -20 & -8 & -71 \\ 3 & -8 & -3 & -29 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1/3 & 1 \\ 0 & 1 & 1/4 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ determined using sympy.ref()

b. $x_1 = 1 + 1/3 x_3$
 $x_2 = 4 - 1/4 x_3$
 x_3 is free

$$\begin{aligned} 4(1) + 12 &= 16 \checkmark \\ 9(5) - 20(1) - 8(12) &= \\ 45 - 20 - 96 &= -71 \checkmark \\ 3(5) - 8(1) - 3(12) &= \\ 15 - 8 - 36 &= -29 \checkmark \end{aligned}$$

c. $x_1 = 1 + 1/3(12) = 5$
 $x_2 = 4 - 1/4(12) = 1$
 $x_3 = 12$

1.62 $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

1.69 $x_1 = -6 + 6x_3 + 2x_5 \Rightarrow 2x_5 = 6 + x_1 - 6x_3$
 $\Rightarrow x_5 = 3 + 1/2 x_1 - 3x_3$

$$\begin{aligned} x_2 &= 4 + 4x_3 + 6(3 + 1/2 x_1 - 3x_3) \\ &= 4 + 4x_3 + 18 + 3x_1 - 18x_3 \\ &= 22 + 3x_1 - 14x_3 \end{aligned}$$

$$\begin{aligned} x_4 &= -4 + 5(3 + 1/2 x_1 - 3x_3) \\ &= -4 + 15 + 5/2 x_1 - 15x_3 \\ &= 11 + 5/2 x_1 - 15x_3 \end{aligned}$$

x_1 is free
 $x_2 = 22 + 3x_1 - 14x_3$
 x_3 is free
 $x_4 = 11 + 5/2 x_1 - 15x_3$
 $x_5 = 3 + 1/2 x_1 - 3x_3$

It's also possible to do this by defining x_3 in terms of x_1 and x_5 and leaving x_5 free.

1.71 That we have distinct planes means their equations are not scalar multiples, so the second row of the RREF is not all zeros. This means the RREF has exactly 2 pivots. This leaves the following possibilities:

$$\begin{bmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 0 & 1 & b \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & a \\ 0 & 0 & 1 & b \end{bmatrix}^*$$

The second two are not possible because this implies the z -coordinate of every solution is the same and that's not the case here. So we're left with the first case. This has the general form solution:

$$\begin{aligned} x &= b - az \\ y &= d - cz \\ z &\text{ is free} \end{aligned}$$

$$* \text{ Note that } \begin{bmatrix} + & + & \neq 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

When $z = 0$ we have

$$\begin{aligned} x &= b = 4 \\ y &= d = 1 \end{aligned}$$

When $z = 1$ we have

is not a possibility because the system is inconsistent.

↳ (next page)

$$x = b - a(1) = 4 - a = 2$$

$$y = d - c(1) = 1 - c = 4$$

So $a = 2$, $b = 4$, $c = -3$, $d = 1$ and the RREF is

$$\begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & -3 & 1 \end{bmatrix}$$